



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Tran Van Son
Oct 1, 2025



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Project methodologies

- **Data Collection:** Used APIs and web scraping to gather SpaceX launch data.
- **Data Wrangling & Cleaning:** Performed data cleaning and transformation to prepare datasets for analysis.
- **Exploratory Data Analysis (EDA):** Visualized and explored data distributions, relationships, and trends
- **Interactive Visual Analytics:** Developed dashboards and interactive visualizations using Plotly Dash
- **Feature Engineering:** Created new features and encoded categorical variables for modeling.
- **Predictive Modeling:** Built and compared machine learning models (Logistic Regression, SVM, Decision Tree, KNN) using cross-validation and hyperparameter tuning.

- Project results

- Data was successfully collected and cleaned from SpaceX launches.
- EDA revealed key factors affecting rocket landing success, such as payload mass, launch site, and orbit.
- Feature engineering enabled effective modeling by converting categorical data and selecting relevant features.
- All tested machine learning models achieved similar accuracy (~83%) in predicting first stage landing success.
- Confusion matrix showed a tendency for false positives (predicting a landing when it did not occur).
- Interactive dashboards provided visual insights into launch locations and outcomes.
- Findings were presented in a structured report and presentation..

Introduction

Project background and context:

SpaceX aims to reduce launch costs by reusing rocket first stages. Predicting whether a rocket will successfully land is crucial for cost estimation and competitive bidding. The project uses real SpaceX launch data to build a predictive pipeline, helping stakeholders understand and forecast landing outcomes.

Problems Statement:

This project aims to determine whether the successful landing of the SpaceX Falcon 9 first stage can be accurately predicted using launch and rocket parameters. The objective is to enhance launch cost estimation, support competitive analysis for alternative launch providers, and support data-driven business and operational decision-making.

Section 1

Methodology

Methodology

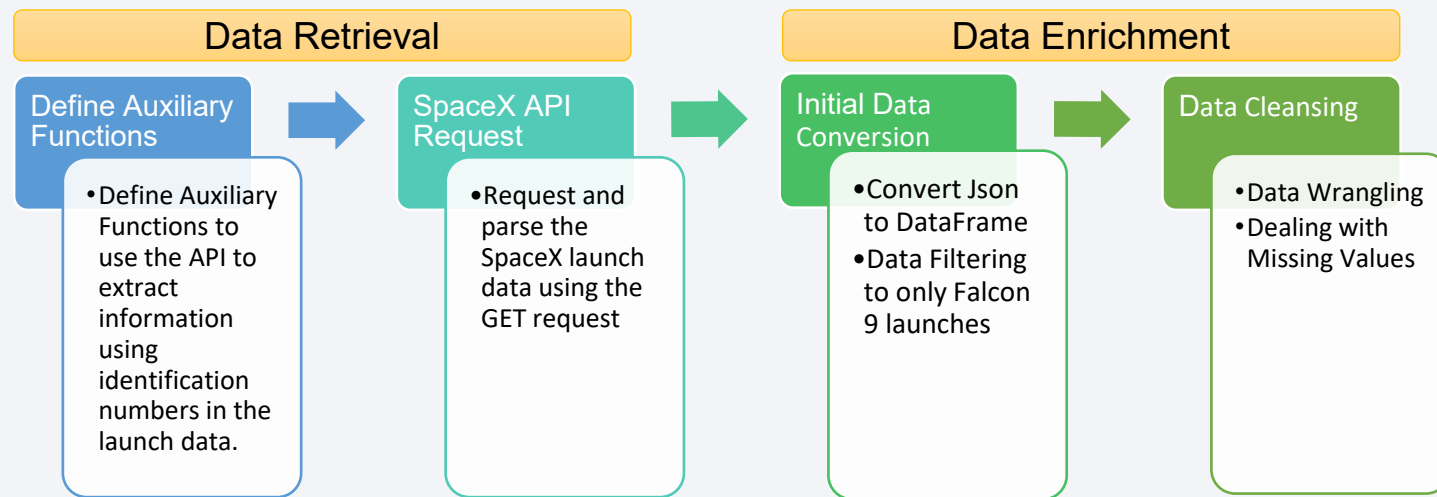
Executive Summary

- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

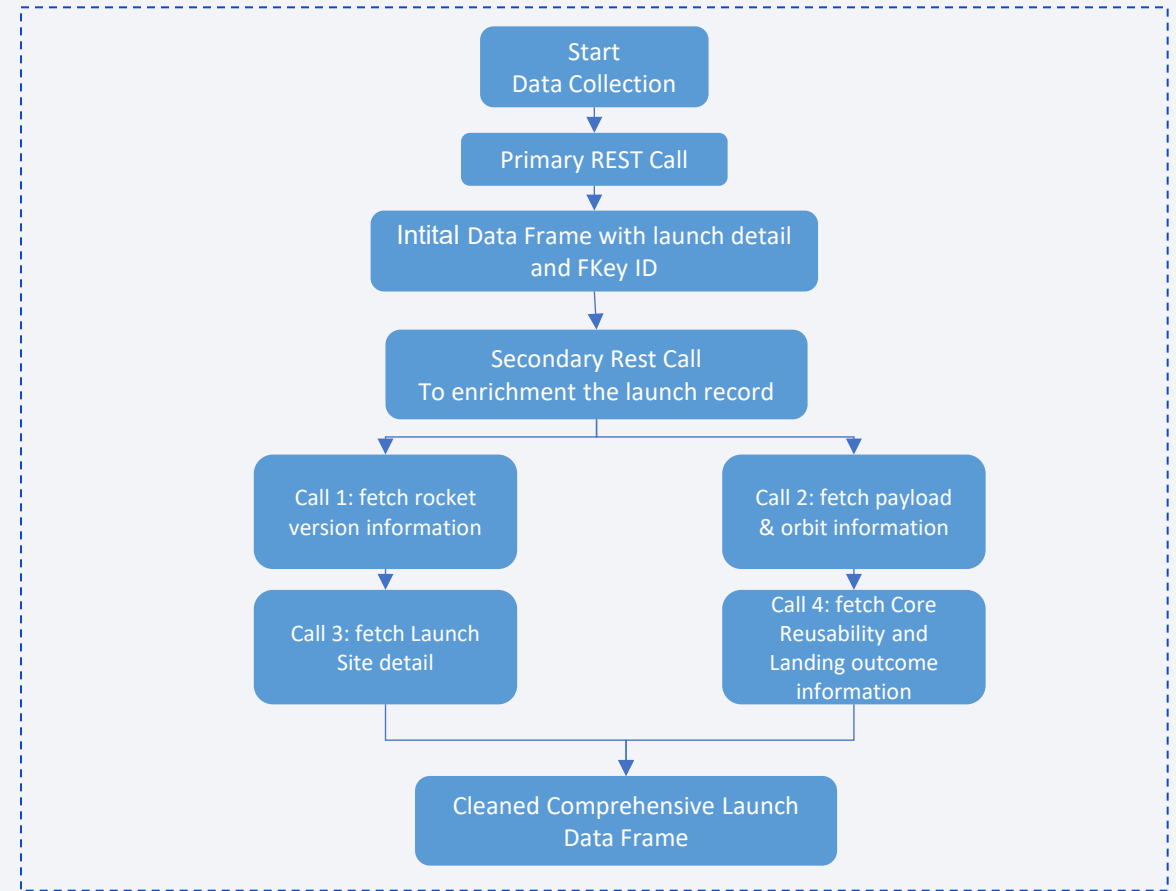
The data set collection process is utilizing the SpaceX API (v4) to gather raw historical launch data, followed by an intensive process of data wrangling to extract and structure hidden details.

Data Collection process and flowcharts



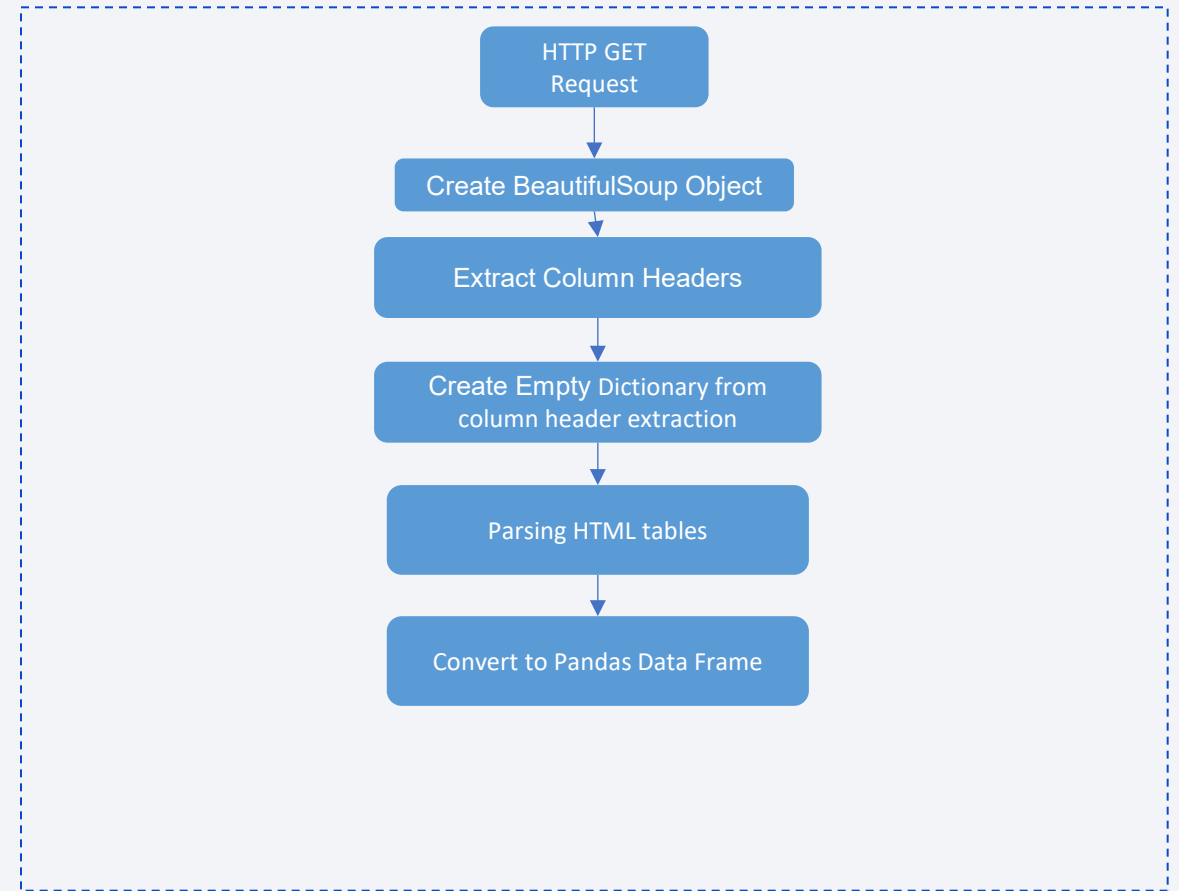
Data Collection – SpaceX API

- Data Collection Process with SpaceX REST Calls:
- The collection process relies on a core principle: **Initial retrieval of launch summaries** followed by **iterative API enrichment** using the extracted IDs.
 - Fetch Core Launch Records to retrieve a list of all historical launches via primary API call which include Foreign Keys ID of Rocket, launchpad, payloads and cores.
 - Fetch detailed metadata (Resolve Rocket Version, Launch Site Detail, Payload & Orbit, Core Reusability and Landing outcome) via four secondary API calls per launch to fill in the missing details.
- GitHub URL:
<https://github.com/transon2772002/TestRepo/blob/Data-Science-Capstone/jupyter-labs-spacex-data-collection-api.ipynb>



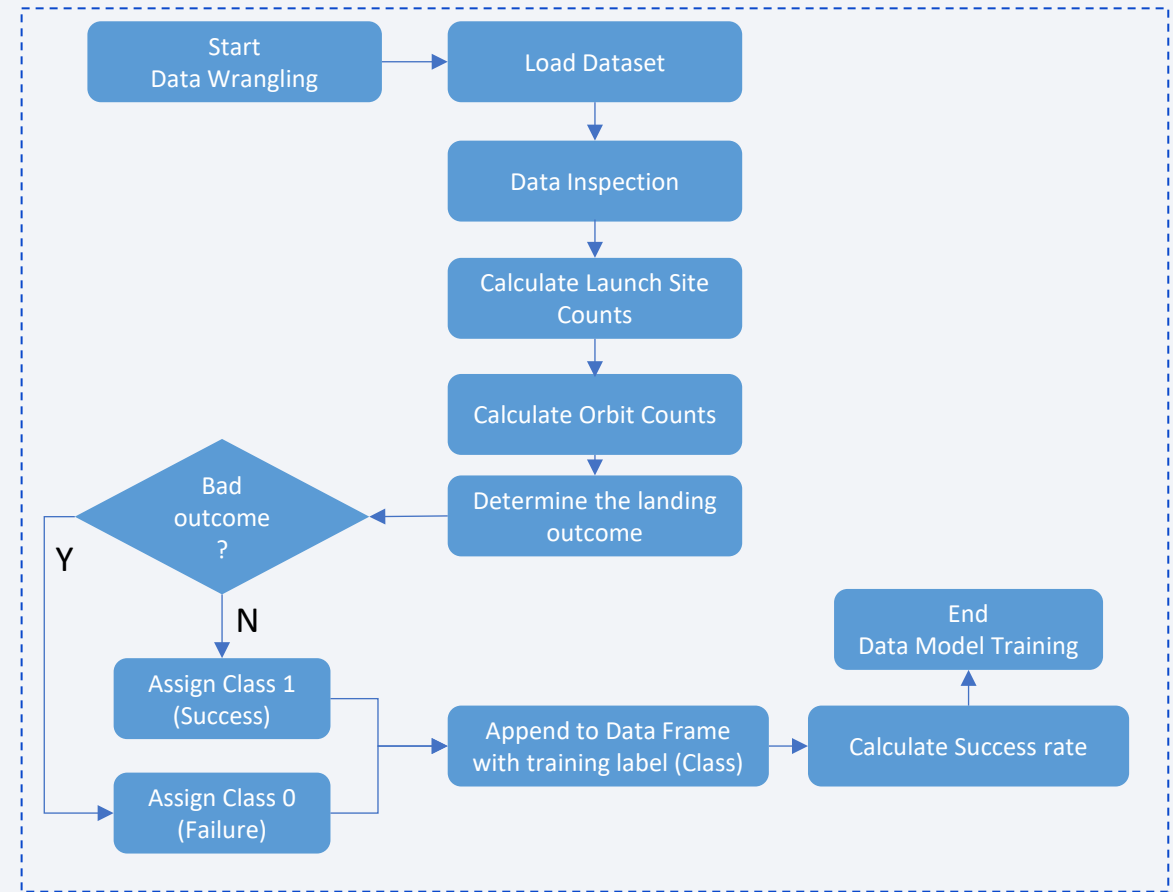
Data Collection - Scraping

- Web scraping process:
 - HTTP GET Request from Falcon9 Launch Wiki page
 - Create BeautifulSoup Object
 - Extract Column Headers
 - Create Empty Dictionary from column header extraction
 - Parsing HTML tables
 - Convert to Pandas DataFrame
- GitHub URL:
<https://github.com/transon2772002/TestRepo/blob/Data-Science-Capstone/jupyter-labs-webscraping%20.ipynb>



Data Wrangling

- The data wrangling process is primarily focused on Exploratory Data Analysis (EDA) and creating the binary classification label required for supervised learning.
 - Load Space X Dataset
 - Identify Missing Values
 - Calculate Launch Site Counts
 - Calculate Orbit Counts
 - Determine Landing Outcomes
 - Define Bad Outcomes Set
 - Create Training Label (Class)
 - Determine the success rate
- The GitHub URL:
<https://github.com/transon2772002/TestRepo/blob/Data-Science-Capstone/labs-jupyter-spacex-Data%20wrangling%20.ipynb>



EDA with Data Visualization

- The exploratory data analysis (EDA) focused on plotting the relationship between key launch parameters and the Class variable (the binary outcome: 1 for successful landing, 0 for failure).
- Scatter Plots of Flight Number vs. Payload Mass: to determine if the two most critical continuous variables, launch experience (FlightNumber) and launch difficulty (PayloadMass), correlate with mission success.
- Scatter Plots of Flight Number vs. Launch Site: To visualize the relationship between Flight Number and Launch Site and investigate if success rates vary significantly based on the Launch Site.
- Scatter Plots of Launch Site vs. Payload Mass: To visualize the relationship between Payload Mass and Launch Site and observe if there is any relationship between launch sites and their payload mass.
- Bar Plots of Success rate of each orbit type: to identify which orbits have the highest success rates.
- Scatter Plots of Flight Number vs. Orbit type: to see if there is any relationship between FlightNumber and Orbit type.
- Scatter Plots of Payload Mass and Orbit type: to see the relationship between Payload Mass and Orbit type
- Line Plot of Average Success Rate and Year: to get the average launch success trend.
- The GitHub URL: <https://github.com/transon2772002/TestRepo/blob/Data-Science-Capstone/edadataviz.ipynb>

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
 1. Unique Launch Sites: Retrieved the distinct names of all launch sites used in the space missions.
 2. Launch Site Filter: Displayed the first 5 records where the Launch_Site begins with 'CCA'
 3. Total Payload Mass (Customer): Calculated the sum of Payload_Mass__KG_ for all missions launched for the customer 'NASA (CRS)'.
 4. Average Payload Mass (Booster): Calculated the average Payload_Mass__KG_ carried by missions using the booster version 'F9 v1.1'.
 5. First Successful Ground Landing: Found the earliest Date on which a 'Success (ground pad)' landing outcome was achieved.
 6. Drone Ship Success Boosters: Listed the Booster_Version names that achieved a 'Success (drone ship)' outcome with a payload mass between 4000 KG and 6000 KG.
 7. Mission Outcome Counts: Grouped the records by Mission_Outcome to count the total occurrences of each outcome (Success, Failure, etc.).
 8. List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.
 9. List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015
 10. Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.
- The GitHub URL: [https://github.com/transon2772002/TestRepo/blob/Data-Science-Capstone/jupyter-labs-eda-sql-coursera_sqllite%20\(1\).ipynb](https://github.com/transon2772002/TestRepo/blob/Data-Science-Capstone/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb)

Build an Interactive Map with Folium

Summary of Map Objects Created

- 1. Launch Site Circles (folium.Circle): A Circle object was created for each of the four main launch sites.
- Purpose: The circles serve to visually highlight the exact geographic area of the launch site on the map.
- 2. Launch Site Markers (folium.map.Marker with DivIcon): A custom Marker object was created for each launch site
- Purpose: The markers provide a textual label for easy identification of the launch sites on the map.
- 3. Launch Outcome Markers (Cluster Markers): Small Marker objects are created for every single launch (not just the site locations) and are then added to a Marker Cluster.
- Purpose: The Cluster Markers to easily identify which launch sites have relatively high success rates
- 4. Draw a PolyLine and distance between a launch site to the selected point folium.PolyLine
- Purpose: PolyLine provide the proximity from launch site to railways, highways, coastline and city

The GitHub URL: https://github.com/transon2772002/TestRepo/blob/Data-Science-Capstone/lab_jupyter_launch_site_location1.ipynb

Build a Dashboard with Plotly Dash

- The SpaceX Launch Records Dashboard using the Dash framework, featuring two main plots and two interactive elements, it creates an interactive dashboard that allows users to quickly analyze SpaceX launch data from multiple perspectives:
- Plots/Graphs:
 - Success Pie Chart (success-pie-chart): Displays the total count of successful (class = 1) versus failed (class = 0) launches.
 - Payload-Success Scatter Chart (success-payload-scatter-chart): Visualizes the relationship (correlation) between the Payload Mass (kg) (x-axis) and the Launch Outcome (class, y-axis, where 1=Success, 0=Failure).
- Interactions (Input Components):
 - Launch Site Dropdown (site-dropdown): Allows to select either 'ALL' Sites or one of the four specific launch sites: 'CCAFS LC-40', 'VAFB SLC-4E', 'KSC LC-39A', or 'CCAFS SLC-40'.
 - Payload Range Slider (payload-slider): Allows to select a minimum and maximum payload mass (kg) range (from 0 to 10,000 kg with steps of 1,000 kg).
- GitHub URL: <https://github.com/transon2772002/TestRepo/blob/Data-Science-Capstone/spacex-dash-app.py>

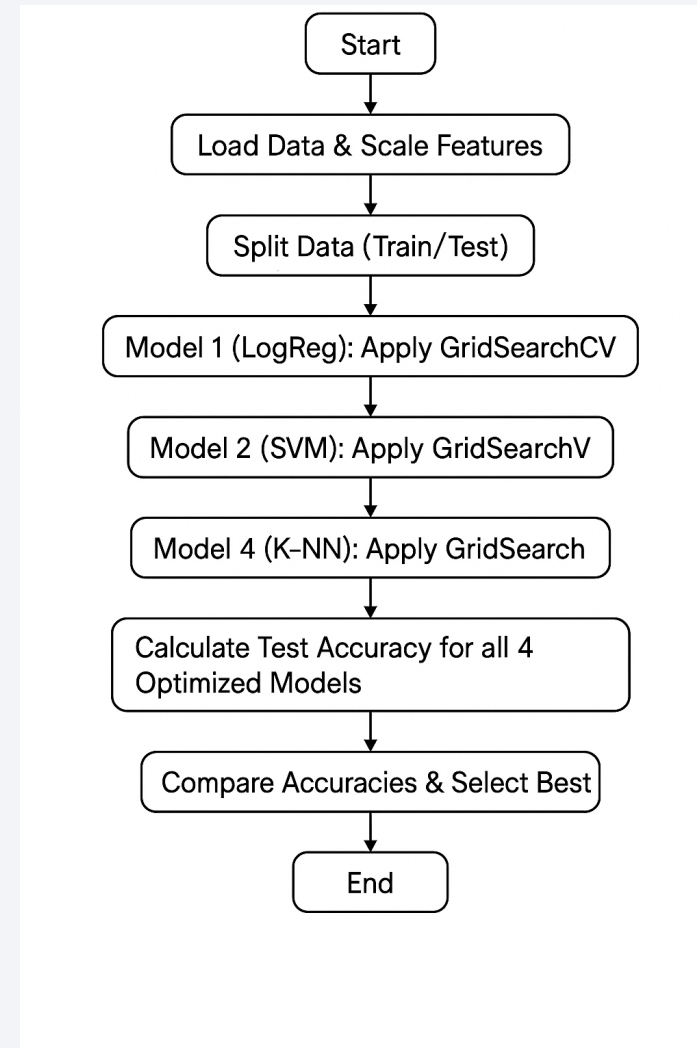
Predictive Analysis (Classification)

Predictive Analysis Summary:

1. **Built:** Four models—Logistic Regression, SVM, Decision Tree, and K-NN—were built using **GridSearchCV** to automatically tune the hyperparameters for each algorithm.
2. **Evaluated:** The models were evaluated using their respective `best_estimator_` (from GridSearchCV) on an independent **test set** to calculate the classification **accuracy score**.
3. **Improved:** Hyperparameter tuning using **GridSearchCV** was the primary method of improvement, systematically searching for the optimal parameters to maximize model performance.
4. **Found the Best:** The "best" model was determined by comparing the final **test accuracy scores**. Since all four models achieved the same highest accuracy (0.8333), they are considered equally best-performing under this metric.

Github URL: https://github.com/transon2772002/TestRepo/blob/Data-Science-Capstone/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

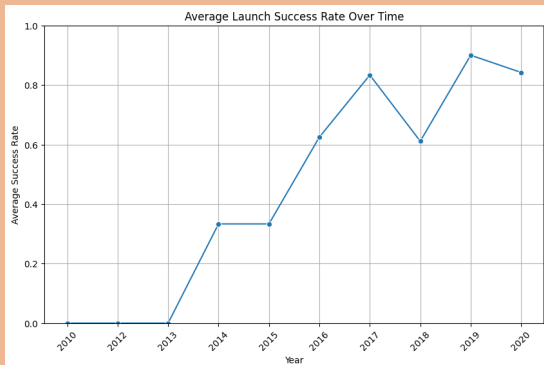
Model development process Flowchart



Results

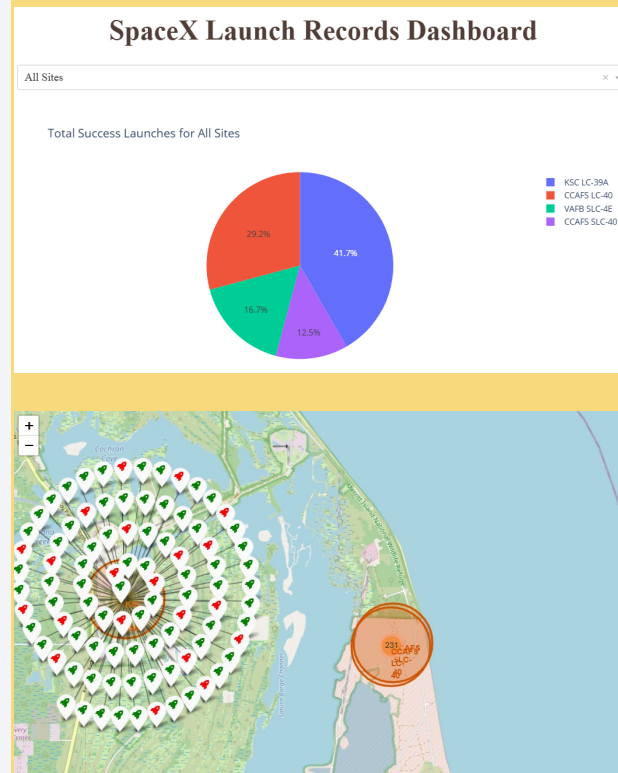
Exploratory data analysis

- Flight Number vs. Payload Mass:** Higher flight numbers and heavier payloads are generally associated with more successful landings.
- Flight Number vs. Launch Site:** Success rates vary by launch site, with some sites showing more successful landings as flight numbers increase.
- Payload Mass vs. Launch Site:** Certain launch sites (e.g., VAFB-SLC) do not launch heavy payloads.
- Success Rate by Orbit Type:** Orbits like ES-L1, GEO, HEO, and SSO have a 100% success rate.
- Flight Number vs. Orbit:** Success in LEO orbit improves with more flights
- Payload Mass vs. Orbit:** Polar, LEO, and ISS orbits have higher success rates for heavy payloads
- Yearly Success Trend:** Launch success rates have increased steadily since 2013.



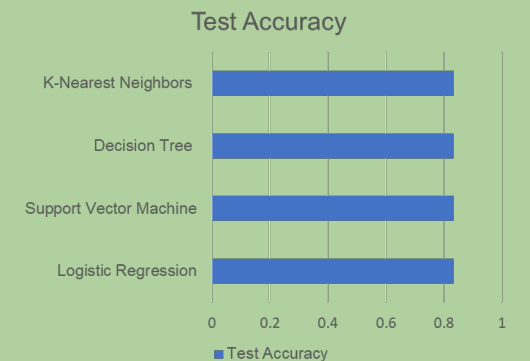
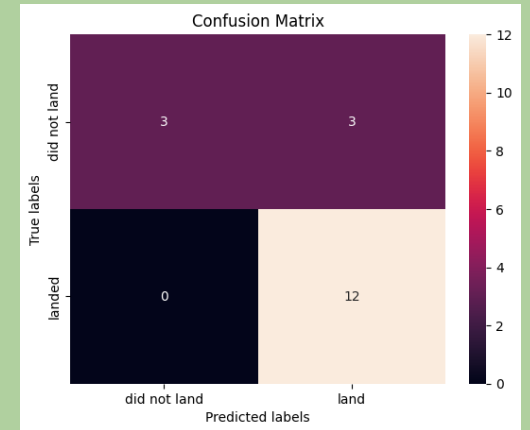
Interactive analytics

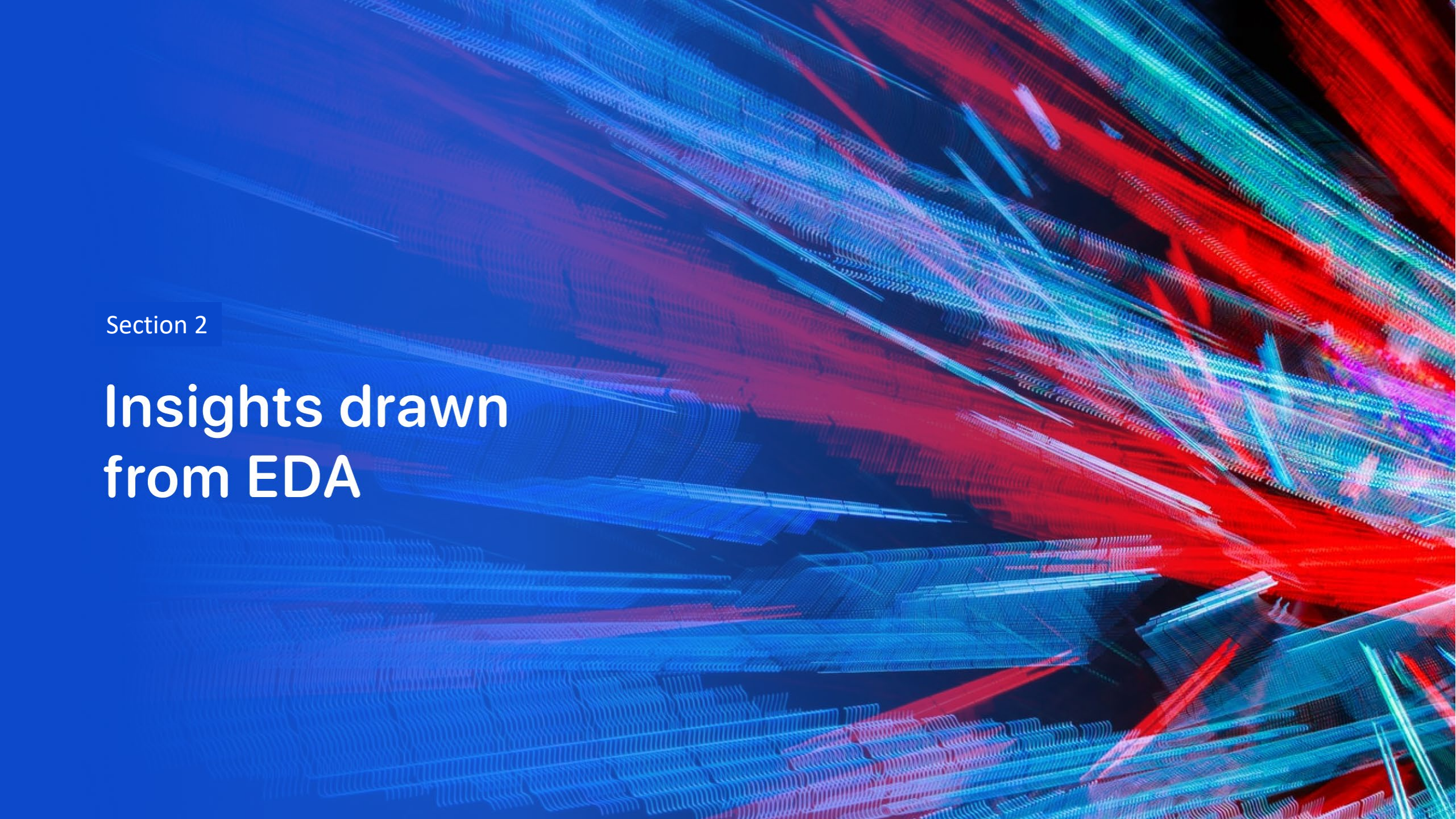
KSC LC 39A (East Coast) demonstrate a high success rate than the rest of the launch site.



Predictive analysis

All four models (Logistic Regression, SVM, Decision Tree, KNN) achieved the same test accuracy of 83.33% on the test set (18 samples).



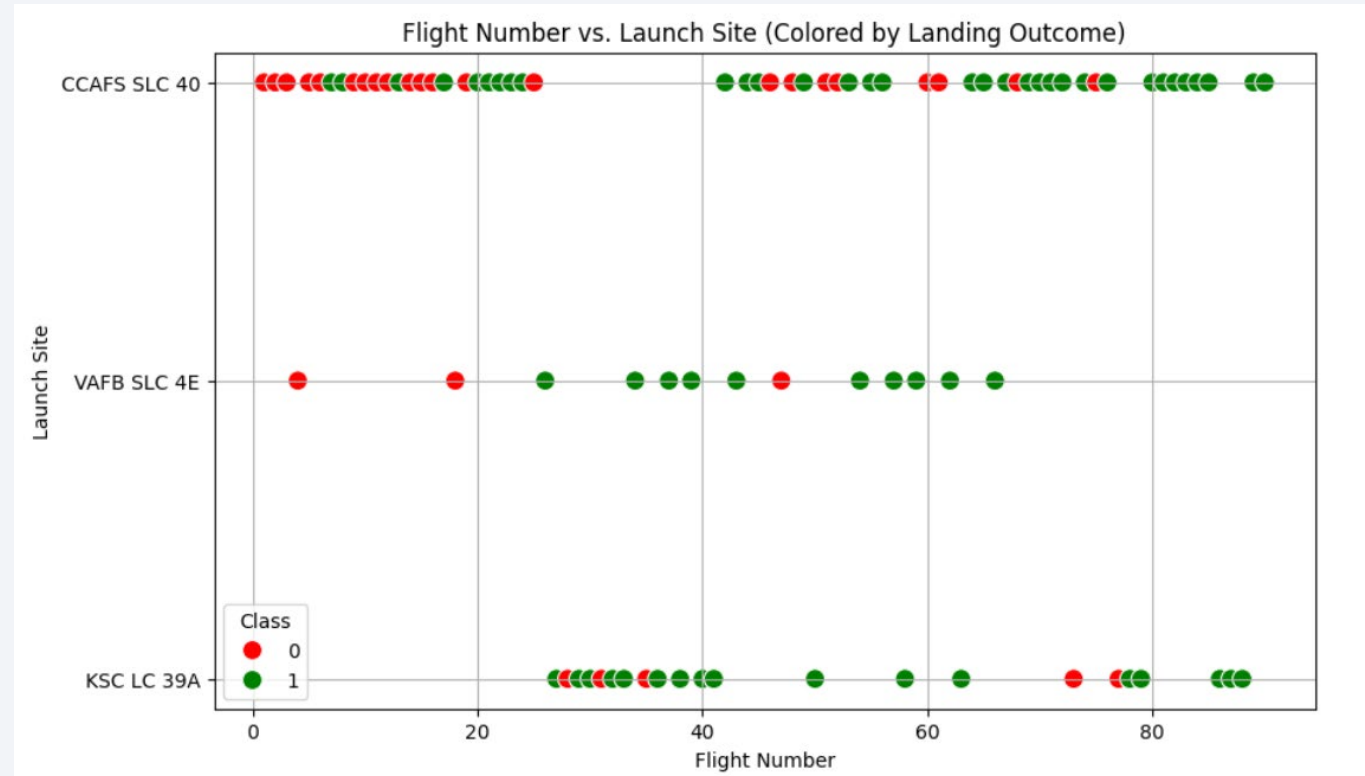
The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- Show a scatter plot of Flight Number vs. Launch Site
- X-axis: Flight Number (0 to ~90)
- Y-axis: Launch Site
 - CCAFS SLC 40
 - VAFB SLC 4E
 - KSC LC 39A
- Color Coding:
 - Red (Class 0): Unsuccessful landing
 - Green (Class 1): Successful landing
- Show the screenshot of the scatter plot with explanations

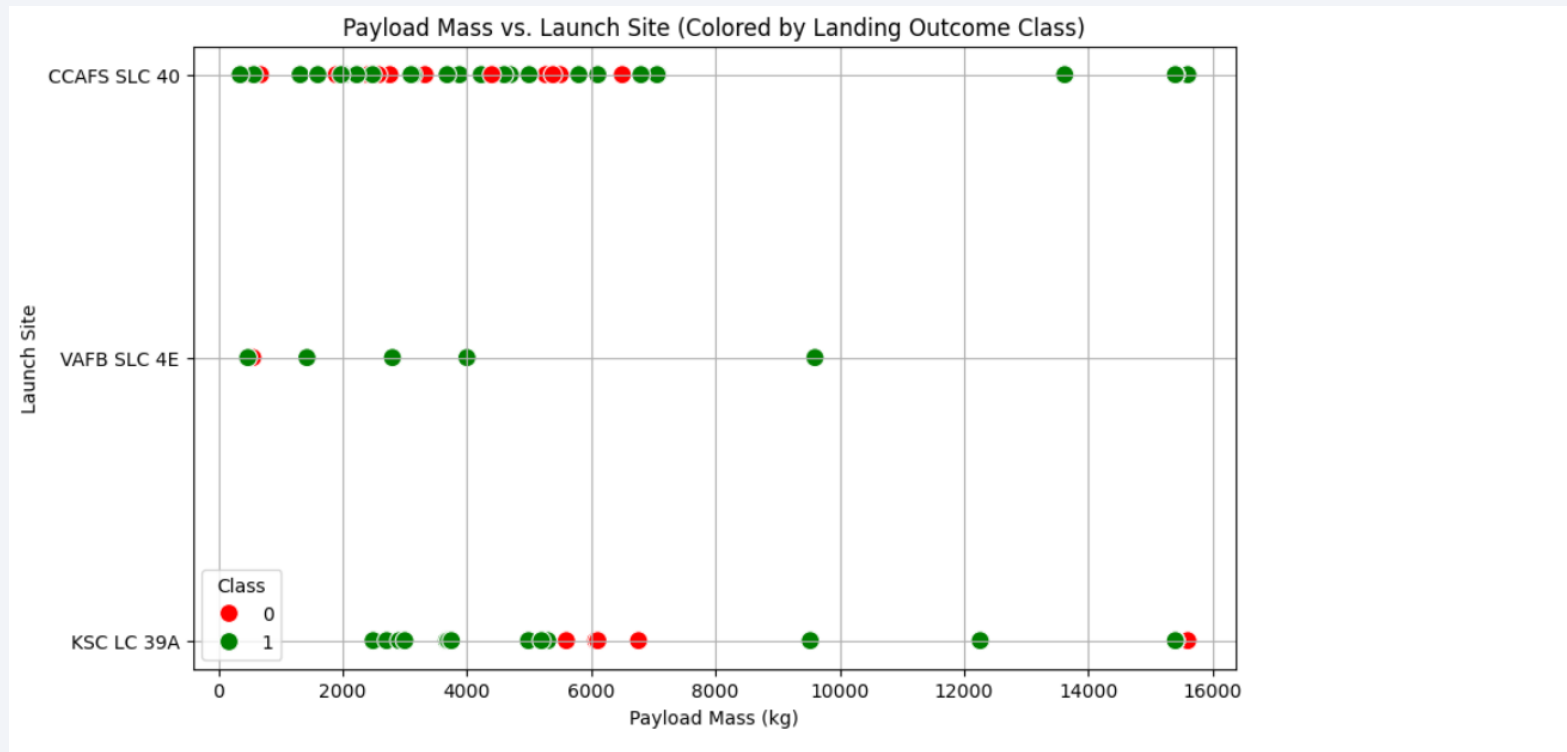


Now try to explain the patterns you found in the Flight Number vs. Launch Site scatter point plots.

The earlier mission shows a mix of red and green which shows the less consistent outcome in early stages. However, the most recent launches across all sites are mostly green, indicating increase success rate over time. The launch site VAFB SLC 4E may be benefited from lesson learned at other sites, had fewer mission with high success rate. Overall, The transition from red to green across all site reflects SpaceX learning curve and interative improvement overtime.

Payload vs. Launch Site

- Show a scatter plot of Payload vs. Launch Site
- X-axis: Payload Mass (kg), ranging from 0 to 16,000 kg.
- Y-axis: Launch Sites:
 - CCAFS SLC 40
 - VAFB SLC 4E
 - KSC LC 39A
- Color Coding:
 - Red (Class 0): Unsuccessful landing
 - Green (Class 1): Successful landing
- Show the screenshot of the scatter plot with explanations

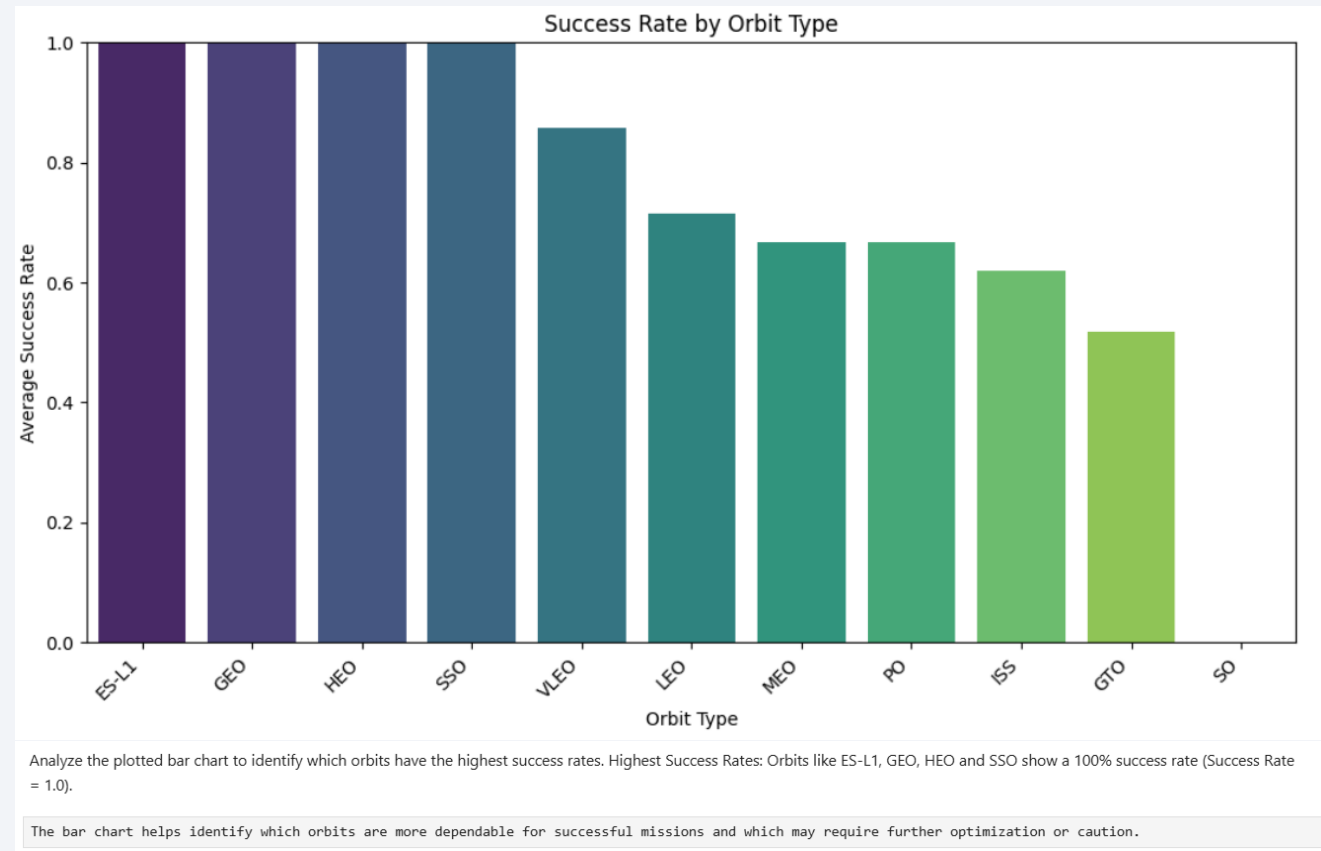


Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

Landing success is not strongly dependent on payload mass, Unsuccessful landing (red) are scattered, with no strong correlation to payload mass and all launch sites have achieved success across a wide range of payloads.

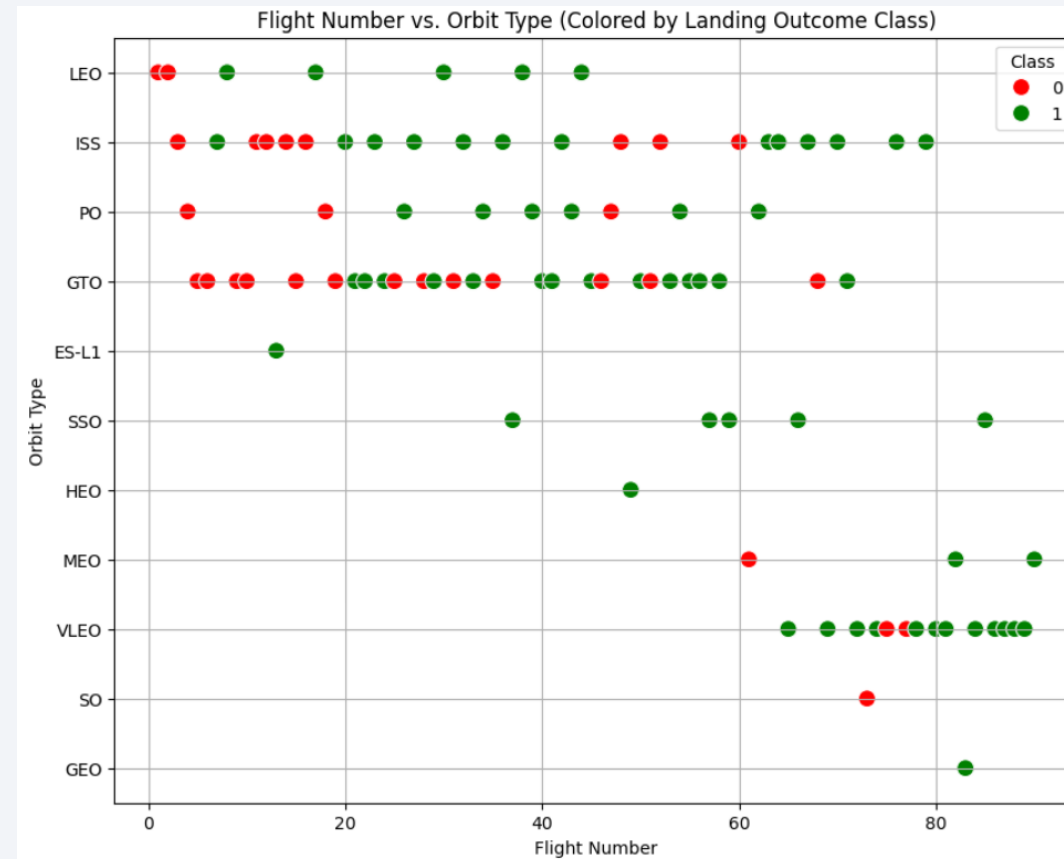
Success Rate vs. Orbit Type

- Show a bar chart for the success rate of each orbit type
- Y-axis: Average Success Rate (from 0.0 to 1.0)
- X-axis: Orbit Types:
- ES-L1, GEO, HEO, SSO, VLEO, LEO, MEO, PO, ES, GTO, SO
- Color Gradient: Bars are shaded from dark purple to light green, likely representing increasing success rates.
- Show the screenshot of the scatter plot with explanations



Flight Number vs. Orbit Type

- Show a scatter point of Flight number vs. Orbit type
- X-axis: Flight Number (0 to ~90)
- Y-axis: Orbit Types:
 - LEO, ISS, PO, GTO, ES-L1, SSO, HEO, MEO, VLEO, SO, GEO
- Color Coding:
 - Red (Class 0): Unsuccessful landing
 - Green (Class 1): Successful landing
- Show the screenshot of the scatter plot with explanations



You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

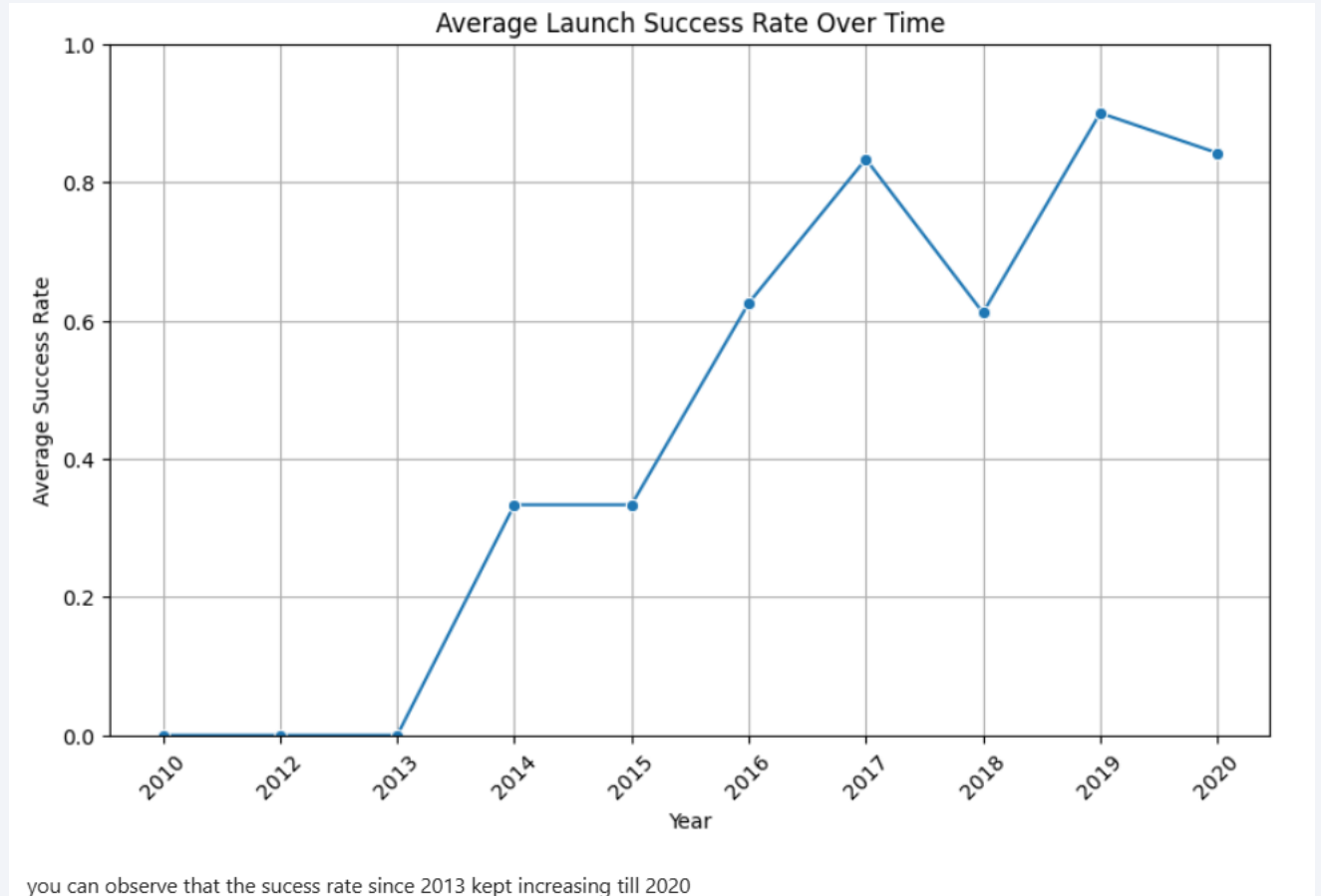
Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type
- X-axis: Payload Mass (kg), ranging from 0 to 16,000 kg.
- Y-axis: Orbit Types:
 - LEO, ISS, PO, GTO, ES-L1, SSO, HEO, MEO, VLEO, SO, GEO
- Color Coding:
 - Red (Class 0): Unsuccessful landing
 - Green (Class 1): Successful landing
- Show the screenshot of the scatter plot with explanations



Launch Success Yearly Trend

- Show a line chart of yearly average success rate
- X-axis: Years from 2010 to 2020
- Y-axis: Average Launch Success Rate, ranging from 0.0 to 1.0
- Line Plot: Connects yearly data points showing how the success rate evolved over time.
- Show the screenshot of the scatter plot with explanations



All Launch Site Names

- Find the names of the unique launch sites
 - CCAFS LC-40
 - VAFB SLC-4E
 - KSC LC-39A
 - CCAFS SLC-40
- Present your query result with a short explanation here
 - Identifies all distinct launch sites used in the dataset.

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE
```

```
* sqlite:///my_data1.db
```

Done.

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with 'CCA'
- Present your query result with a short explanation here
 - Filters and displays the first 5 launches from CCAFS LC-40 sites.

Display 5 records where launch sites begin with the string 'CCA' ⓘ

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
```

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Calculate the total payload carried by boosters from NASA: 45,596 kg
- Present your query result with a short explanation here
 - Total payload mass launched by NASA's CRS.

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
SUM(PAYLOAD_MASS__KG_)
```

```
45596
```

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1:
~2534.67 kg
- Present your query result with a short explanation here
 - Shows the average payload mass carried by the F9 v1.1 booster version.

Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
AVG(PAYLOAD_MASS_KG_)
```

```
2534.6666666666665
```

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad: December 22, 2015
- Present your query result with a short explanation here:
 - Identifies the earliest date of a successful ground landing.

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%sql SELECT MIN(Date) FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
MIN(Date)
```

```
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
 - F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2
- Present your query result with a short explanation here
 - Lists booster versions that meet both landing and payload criteria.

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
 - Success: 99
 - Failure (in flight): 1
 - Success (payload status unclear): 1
- Present your query result with a short explanation here
 - Gives a breakdown of mission outcomes.

List the total number of successful and failure mission outcomes

```
%sql SELECT "Mission_Outcome", COUNT(*) AS Total_Count FROM SPACEXTABLE GROUP BY "Mission_Outcome"
```

```
* sqlite:///my_data1.db
```

Done.

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- Present your query result with a short explanation here
 - Identifies all boosters that carried the heaviest payloads.

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);
```

```
* sqlite:///my_data1.db  
Done.
```

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- Present your query result with a short explanation here
 - Shows months in 2015 when drone ship landings failed.

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%sql SELECT SUBSTR(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE Landing_Outcome = 'Failure (drone ship)' AND SUBSTR(D
```

```
* sqlite:///my_data1.db
```

Done.

	Month	Landing_Outcome	Booster_Version	Launch_Site
	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Present your query result with a short explanation here
 - Shows the number of landing outcome between the date 2010-06-04 and 2017-03-20

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%sql SELECT "Landing_Outcome", COUNT(*) AS Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY Outc
```

* sqlite:///my_data1.db

Done.

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

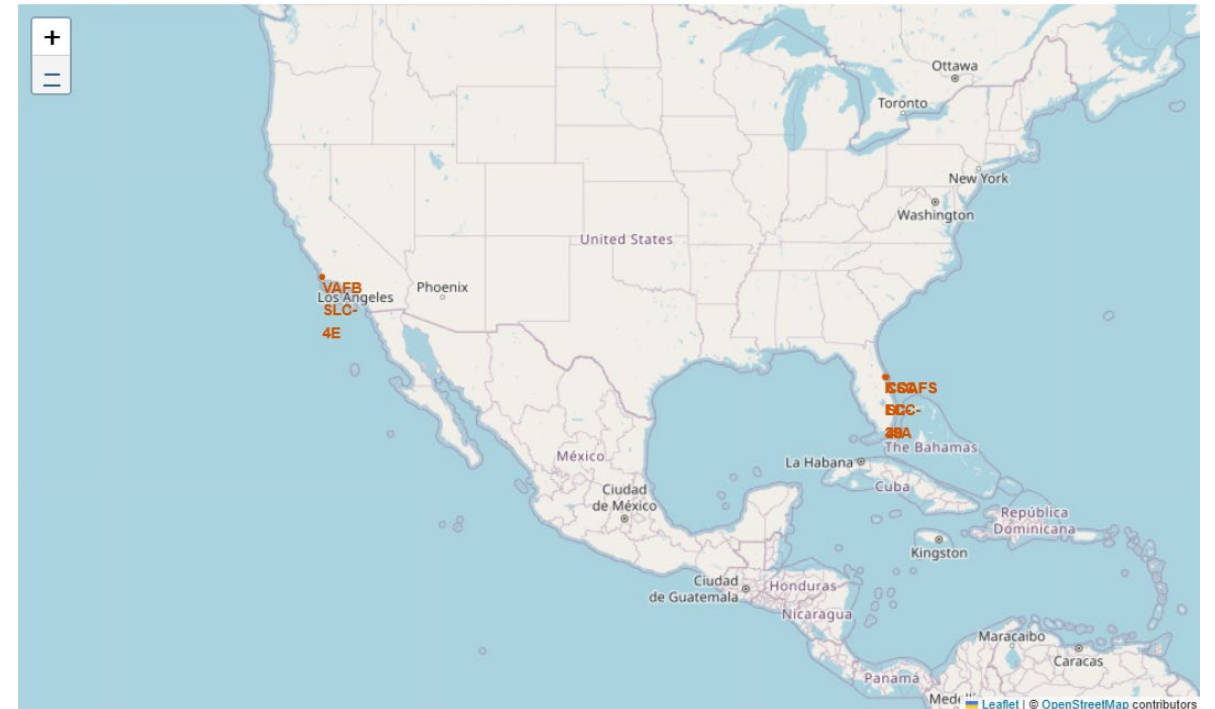
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a curved line separating the dark surface from the deep blue of space.

Section 3

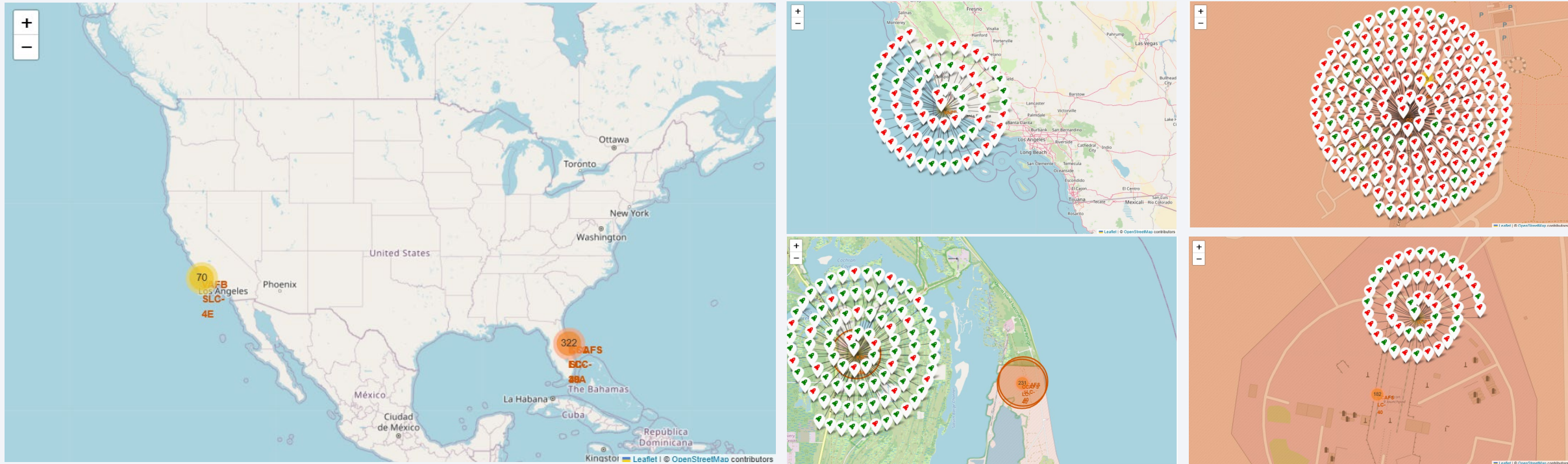
Launch Sites Proximities Analysis

Launch Site Location on Map

- The map clearly marks the geographical locations of these primary launch complexes used by SpaceX
- VAFB SLC 4E (West Coast, near Los Angeles): Used for missions requiring high-inclination or polar orbits. Launches fly south over the Pacific Ocean to avoid populated areas.
- CCAFS & KSC (East Coast, Florida): Used for missions requiring equatorial or low-inclination orbits. Launches fly east over the Atlantic, leveraging the Earth's rotational velocity to gain an efficiency boost.



The success/failed launches for each site on the map



KSC LC 39A (East Coast) and VAFB SLC 4E (West Coast): These sites show a dramatically higher proportion of green markers (successes), demonstrating a high success rate than the rest of the launch site.

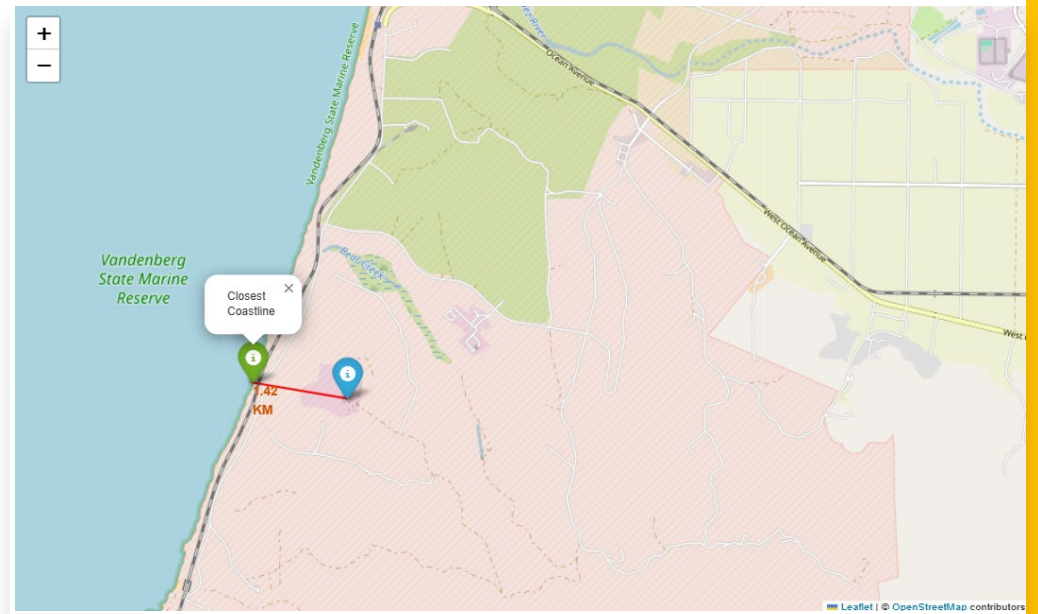
The launch site distances and proximities on the map

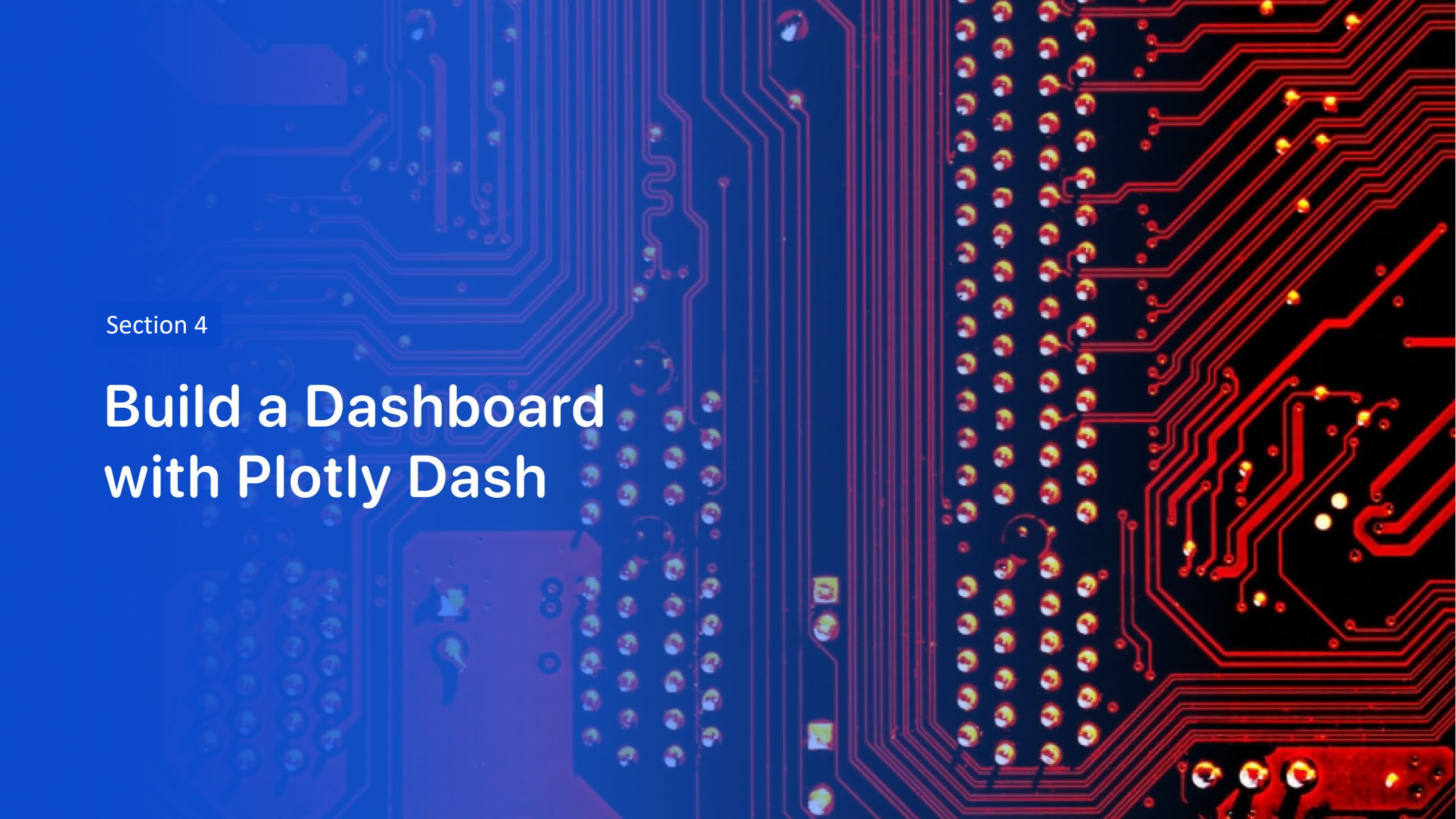
From the launch site's VAFB SLC 4E (West Coast) to Coastline: 1.42 km.

From the launch site's VAFB SLC 4E (West Coast) to Railway: 1.49 km.

From the launch site's VAFB SLC 4E (West Coast) to Closest city: 13.92 km.

From the launch site's VAFB SLC 4E (West Coast) to Highways: 15.97 km.



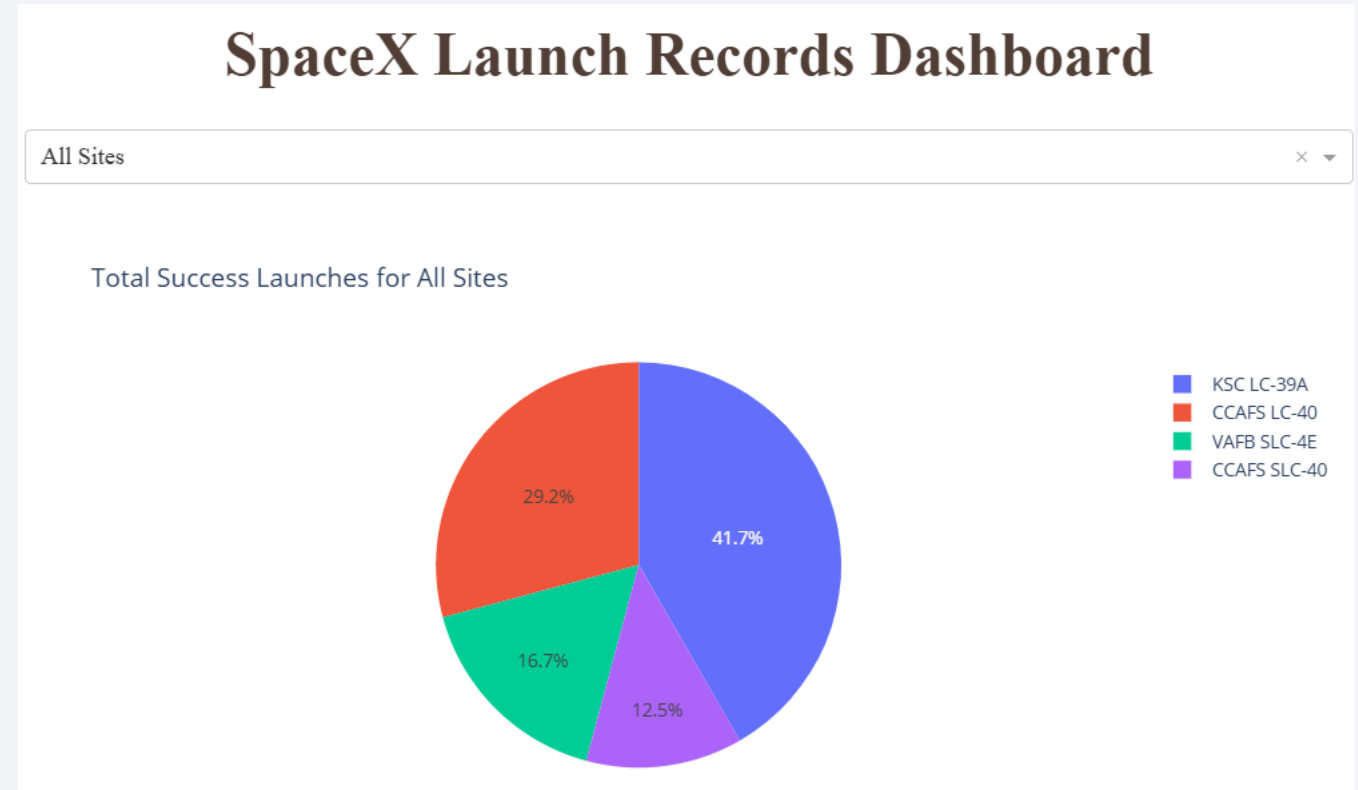


Section 4

Build a Dashboard with Plotly Dash

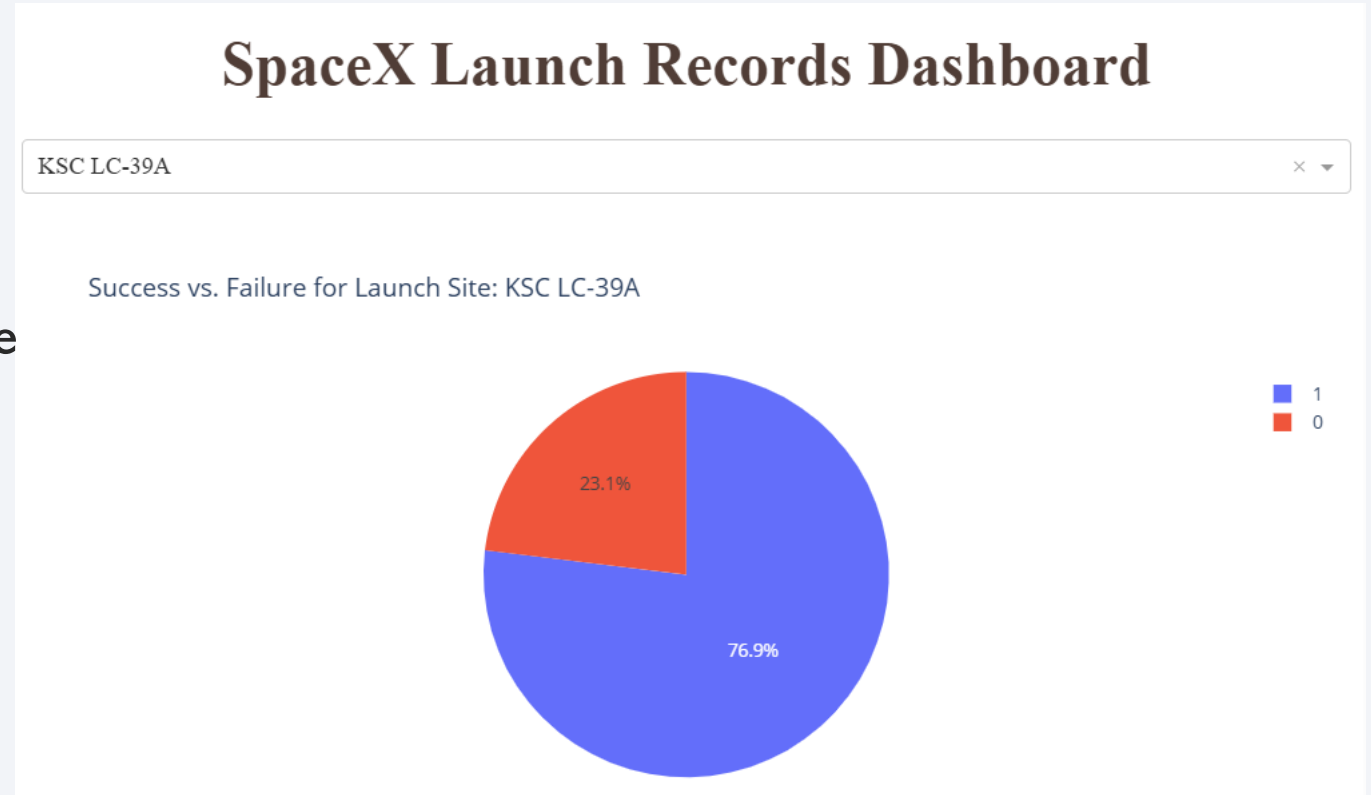
SpaceX Total Success Launch Records Dashboard

- Dashboard Title: SpaceX Launch Record Dashboard
- Launch Site Dropdown: The dropdown menu is visible and currently set to "All Sites."
- Chart Title: The title, "Total Success Launches for All Sites,"
- Pie Chart: This is the primary visualization, it breaks down the total successful launches by the specific Launch Site.
- The pie chart reveals the distribution of all successful SpaceX launches across the four different launch sites, where the highest percentage of Total Successful Launches is the site's KSC LC-39A



SpaceX highest launch success ratio Records Dashboard

- Dashboard Title: SpaceX Launch Record Dashboard
- Launch Site Selection: The dropdown menu is set to KSC LC-39A, meaning the chart only reflects data for launches from this site.
- Chart Title: The title is "Success vs. Failure for Launch Site: KSC LC-39A," confirming the chart's focus.
- Pie Chart: This visualization breaks down the outcomes of all launches from KSC LC-39A into two categories: Success (1) and Failure (0)
- The pie chart reveal that 76.9% of the launches from KSC LC-39A were successful (class 1).



SpaceX Payload vs. Launch Outcome Dashboard

- **Payload Range Slider:** Located at the top, it shows that the current payload filter is set to the full range of 0 Kg to 10,000 Kg. This means the scatter plot is currently displaying all launches within this mass range.
- **Chart Title:** The title, "Correlation between Payload and Success for All Sites," indicates that the data is aggregated across all launch sites.
- **Scatter Plot:** This graph shows the correlation between the two primary variables:
- **X-axis:** Payload Mass (kg), ranging from 0 to 10,000 kg.
- **Y-axis:** Class (Launch Outcome), where 1 represents a Successful Launch and 0 represents a Failure.
- **Color-Coding:** The points are colored based on the Booster Version Category allowing analysis of how different booster types perform across payload.
- The scatter plot reveals clear patterns in the relationship between payload mass and launch outcome for all sites. The red (v1.1) and (FT) boosters account for several of the failures in the lower payload range.



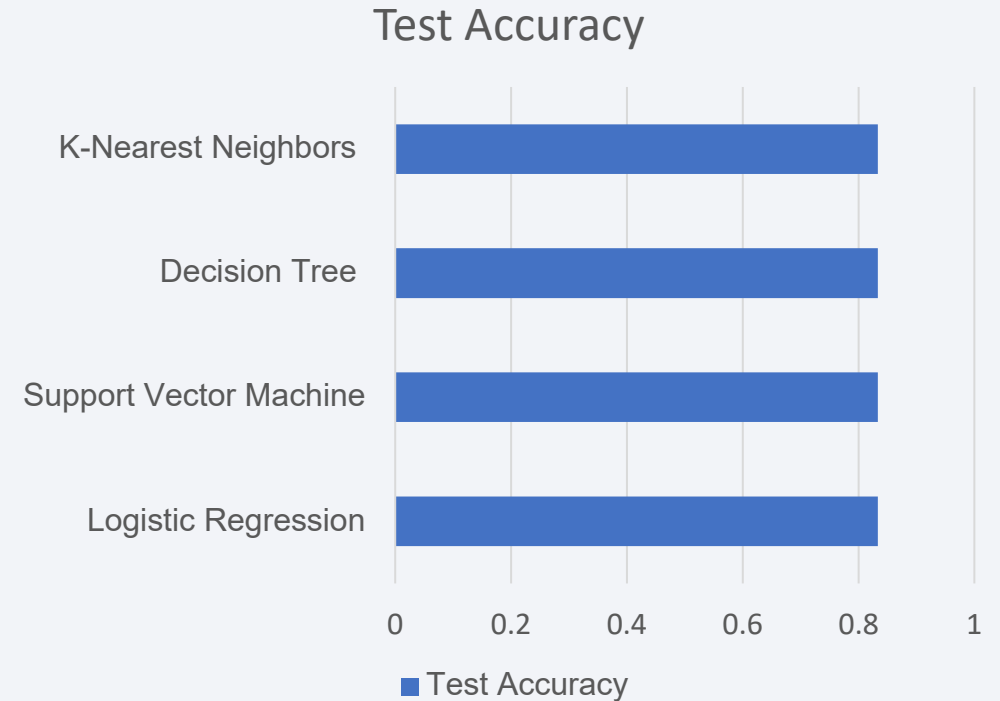


Section 5

Predictive Analysis (Classification)

Classification Accuracy

- The test accuracy values for all four models (logreg_accuracy, svm_accuracy, tree_accuracy, knn_accuracy) are 0.833 (83.33%), is likely a consequence of the very small size of the test set size (18 test samples)

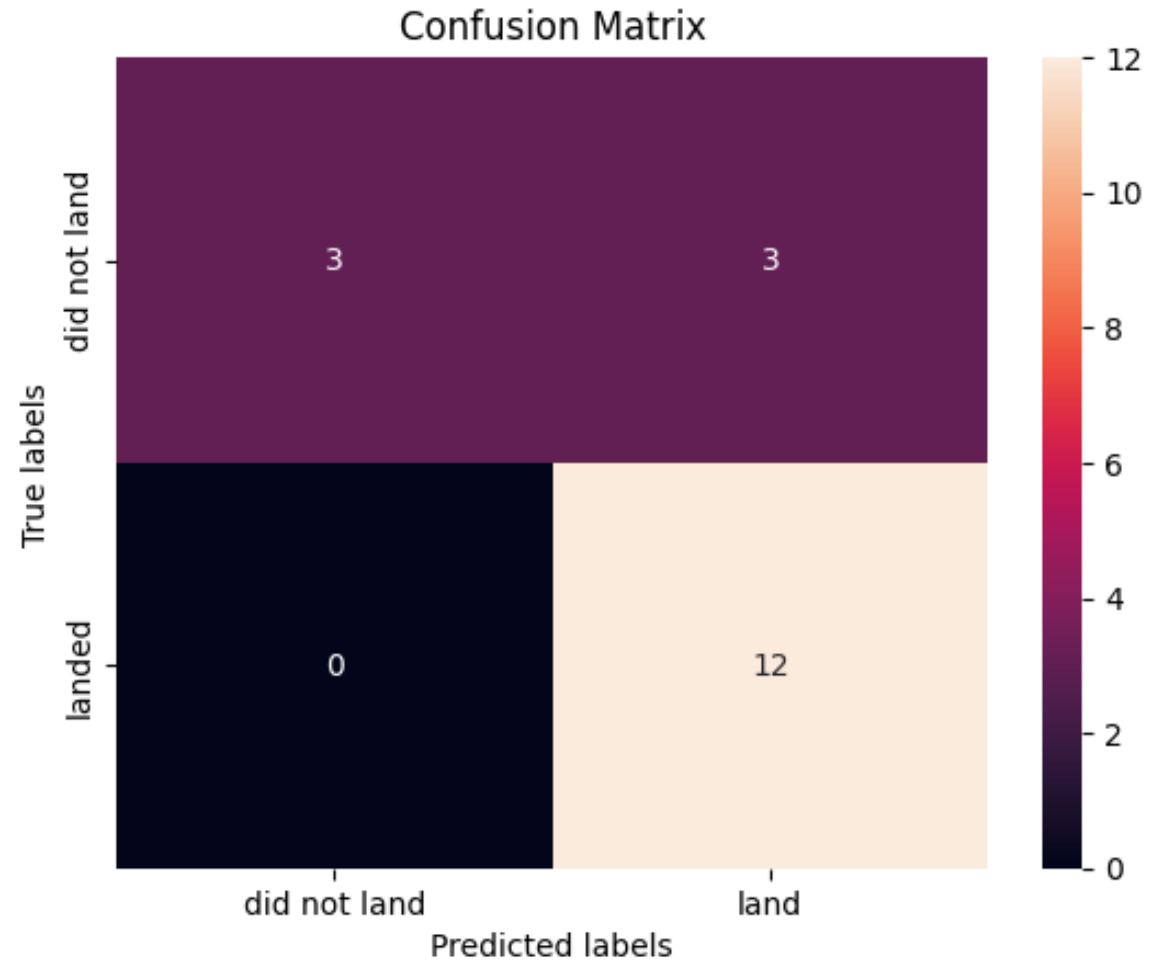


Confusion Matrix

The logistic regression can distinguish between the different classes. We see that the problem is false positives.

True Positive - 12 (True label is landed, Predicted label is also landed)

False Positive - 3 (True label is not landed, Predicted label is landed)



Conclusions

- All four models (Logistic Regression, SVM, Decision Tree, KNN) achieved the same test accuracy of 83.33% on the test set (18 samples).
- The confusion matrix for Logistic Regression shows:
 - True Positives: 12 (correctly predicted landings)
 - False Positives: 3 (predicted landing, but actually did not land)
- The main error type is false positives, meaning the model sometimes predicts a landing when it did not occur.
- All tested machine learning models performed similarly, with moderate accuracy and a common issue observed was the tendency to overpredict landings relative to actual events.. Further tuning or more data may be needed for better performance.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

